

SOC SECURITY MONITORING DASHBOARD

Building a Tier 1 Analyst Monitoring and Triage Dashboard in Splunk

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Active Directory Home Lab

Splunk Enterprise | Sysmon | Windows Security | Windows Firewall | Kali Linux

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1. Project Overview

The goal of this project was to build a centralized SOC monitoring dashboard in Splunk that gives a Tier 1 analyst an at-a-glance view of suspicious authentication, network, and endpoint/process activity. The dashboard uses three Single Value panels to summarize activity matching defined SPL criteria and three corresponding detail tables to provide the context needed for triage and investigation.

The three monitoring areas are Suspicious Authentication Activity, Potential Network Reconnaissance, and Suspicious Process Activity. Rather than presenting large volumes of telemetry at the same visual level, the dashboard follows a Notice -> Prioritize -> Investigate workflow: the Single Values identify how many suspicious occurrences are present within the selected time range, while the detailed tables show the source, account, timing, process, or command-line context associated with those occurrences.

A seven-day global time range was used for the home-lab implementation so previously generated validation activity remained visible during development. In a production SOC, the appropriate monitoring window and refresh interval would depend on shift coverage, telemetry volume, and the organization's triage workflow.

2. Lab Environment

System / Tool	Role
WS01 - 192.168.100.20	Windows 10 Pro domain workstation. Generated Windows Security, Windows Firewall, and Sysmon telemetry used by the dashboard.
DC01 - 192.168.100.10	Windows Server 2025 domain controller hosting Splunk Enterprise and receiving forwarded telemetry.
KALI01 - 192.168.100.30	Kali Linux system used to generate controlled network reconnaissance and failed-authentication activity.
Splunk Enterprise	Used for SPL investigation, dashboard searches, Single Value visualizations, and analyst-focused detail tables.
Splunk Universal Forwarder	Forwarded telemetry from WS01 to Splunk Enterprise on DC01.
Sysmon	Provided process-creation telemetry, including Image, ParentImage, CommandLine, and user context.

3. Dashboard Planning & Design

The dashboard was designed around three analyst questions: Are accounts receiving numerous failed logins in a short timeframe? Is a source IP contacting an unusually large number of destination ports in a short period? Are suspicious PowerShell process-creation events occurring, and what process and user context surrounds them?

Single Value visualizations were selected for the top row because they make the number of matching suspicious occurrences immediately visible. Detailed tables were placed underneath because they preserve the fields required for investigation without overwhelming the analyst with raw telemetry. This separation keeps the dashboard readable while still allowing rapid movement from summary information into event context.

4. Suspicious Authentication Monitoring

For this project, suspicious authentication activity was defined as five or more failed login attempts associated with the same source IP and account within 120 seconds. Windows Security Event ID 4625 provided the failed-logon telemetry. The search groups events by Source_Network_Address and Account_Name, counts the failures, records the first and last failure, calculates the elapsed window in seconds, and retains only groups that meet the threshold.

```
index=main host=WS01 EventCode=4625
| eval Account_Name=mvfilter(Account_Name!="")
| stats count earliest(_time) as First_Failure latest(_time) as Last_Failure by Source_Network_Address Account_Name
| eval Window_Seconds=Last_Failure-First_Failure
| where Window_Seconds <= 120 AND count >= 5
| stats count as Suspicious_Authentication_Activity
```

The Single Value reports the number of source-IP/account groupings that meet the suspicious-authentication criteria. The detail table removes the final summary count and instead displays Source_Network_Address, Account_Name, count, Window_Seconds, First_Failure, and Last_Failure so the analyst can see what account was targeted, where the activity originated, how many failures occurred, and how quickly they occurred.

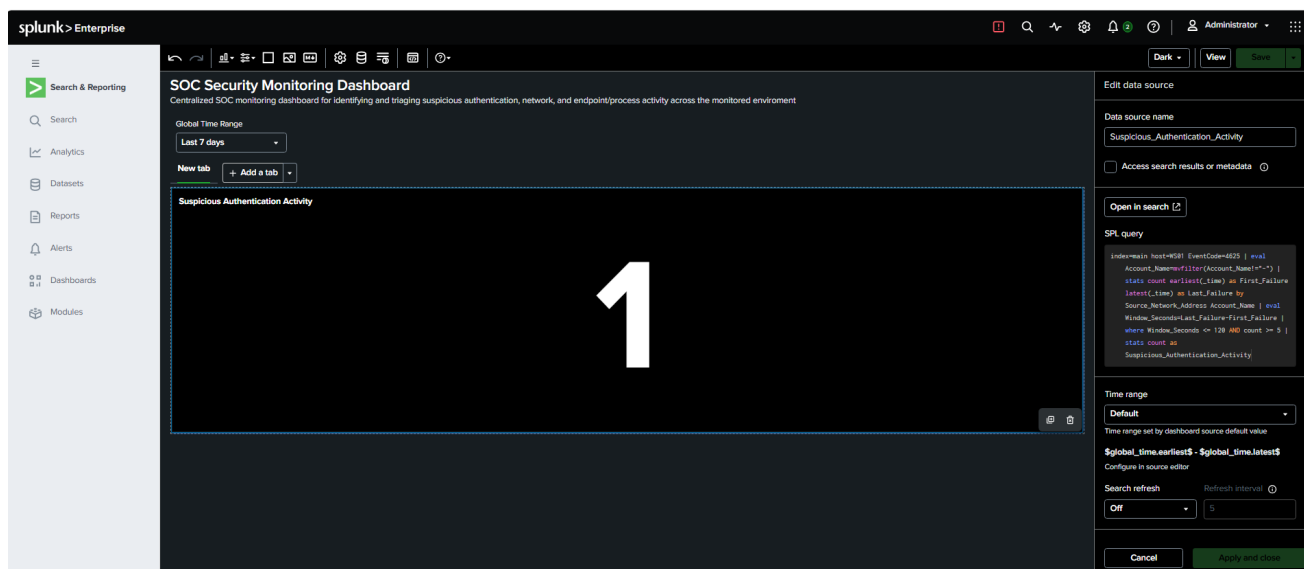


Figure 1 - Suspicious Authentication Activity Single Value used as the top-level authentication indicator.

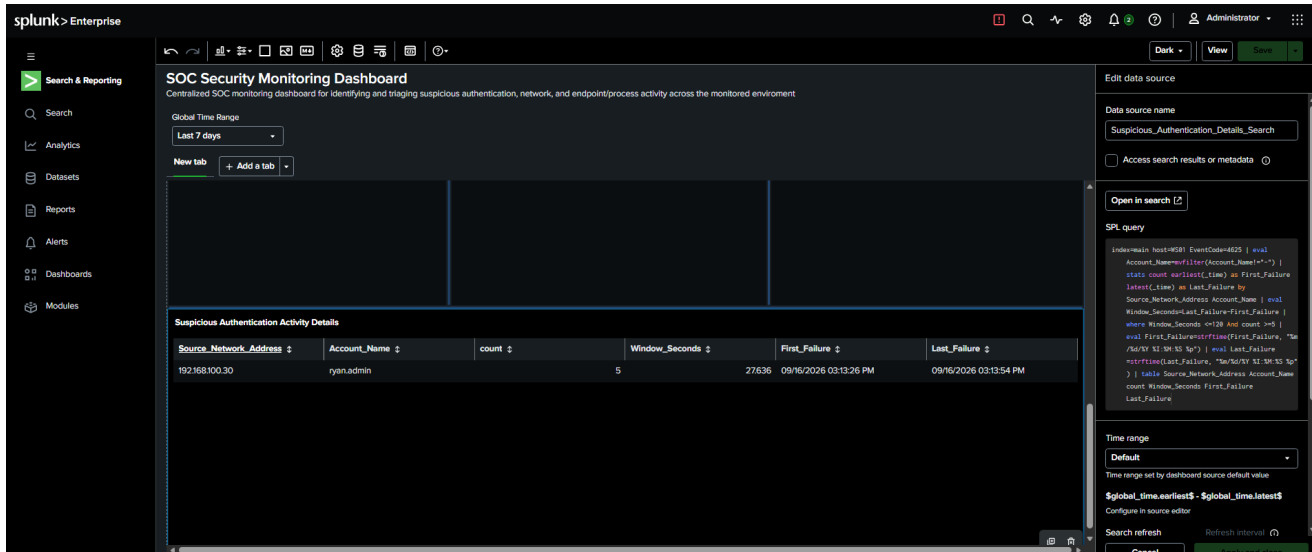


Figure 2 - Suspicious Authentication Activity Details showing the source, account, failure count, and timing context.

5. Potential Network Reconnaissance Monitoring

Potential network reconnaissance was defined as a source IP contacting 100 or more unique destination ports on WS01 within a two-minute window. Windows Filtering Platform Event ID 5152 provided blocked-packet telemetry. A distinct count of Destination_Port was used instead of a simple packet count because packet volume alone does not show how many different ports were targeted. A source can generate many packets against a single service, while contacting a large number of unique ports in a short period is more consistent with port-scanning or reconnaissance behavior.

```
index=main host=WS01 EventCode=5152
| bin _time span=2m
| stats dc(Destination_Port) as Unique_Ports by _time Source_Address
| where Unique_Ports >= 100
| stats count as Potential_Network_Reconnaissance
```

The threshold was tuned using observed home-lab activity. Controlled Nmap scanning from KALI01 produced approximately 998 unique destination ports in a two-minute bucket, while observed background activity was substantially lower. The detail table retains the qualifying time bucket, Source_Address, and Unique_Ports so an analyst can immediately identify the source and scale of the activity.

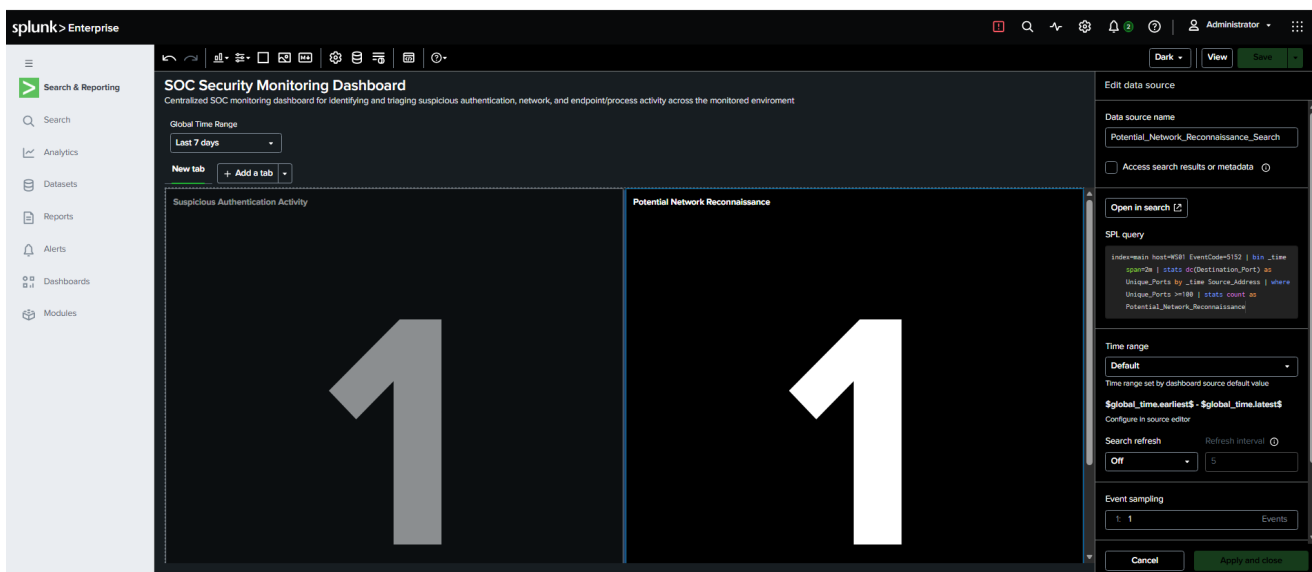


Figure 3 - Potential Network Reconnaissance Single Value summarizing qualifying two-minute source-IP windows.

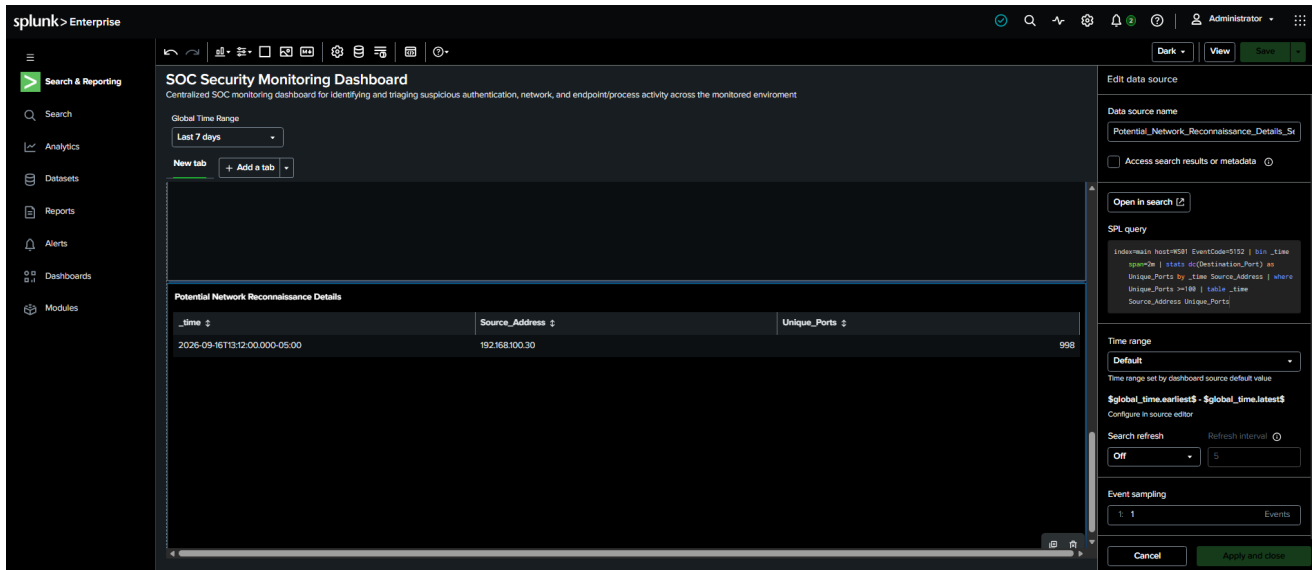


Figure 4 - Potential Network Reconnaissance Details showing KALI01 and 998 unique destination ports.

6. Suspicious Process Activity Monitoring

Suspicious process activity was focused on Sysmon Event ID 1 PowerShell process-creation events where CommandLine contained either -EncodedCommand or the abbreviated -enc form. Encoded PowerShell was treated as suspicious behavior that warrants investigation, not proof of malicious activity.

```
index=main host=WS01 sourcetype=sysmon EventCode=1 Image="*powershell.exe" (CommandLine="*-EncodedCommand*"
OR CommandLine="*-enc*")
| stats count as Suspicious_Process_Activity
```

The detail search preserves _time, host, User, Image, ParentImage, and CommandLine. During development, the User field included a NOT_TRANSLATED value in addition to the useful account value, so mvfilter was used to remove that noise before the results were presented to the analyst.

```
index=main host=WS01 sourcetype=sysmon EventCode=1 Image="*powershell.exe" (CommandLine="*-EncodedCommand*"
OR CommandLine="*-enc*")
| eval User=mvfilter(User!="NOT_TRANSLATED")
| table _time host User Image ParentImage CommandLine
```

These fields provide immediate triage context: Image identifies the process that executed, ParentImage shows what launched it, CommandLine shows how it was executed and what arguments were supplied, User identifies the associated account, and host identifies the affected endpoint. Together they help an analyst decide whether additional investigation is required.

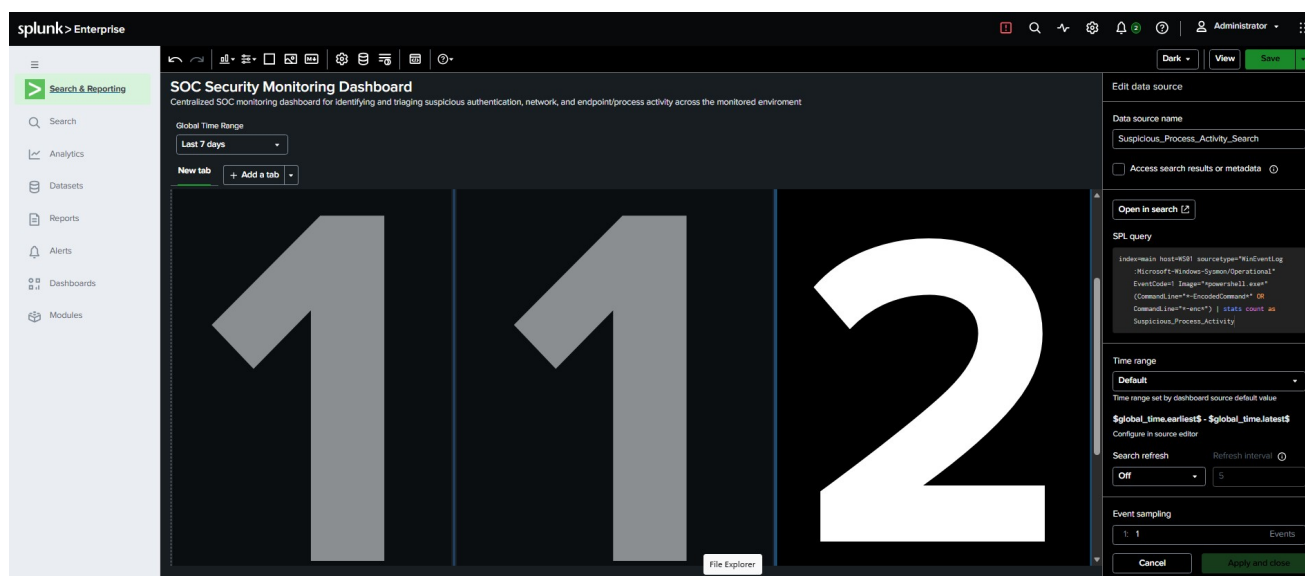


Figure 5 - Suspicious Process Activity Single Value summarizing matching encoded PowerShell process-creation events.

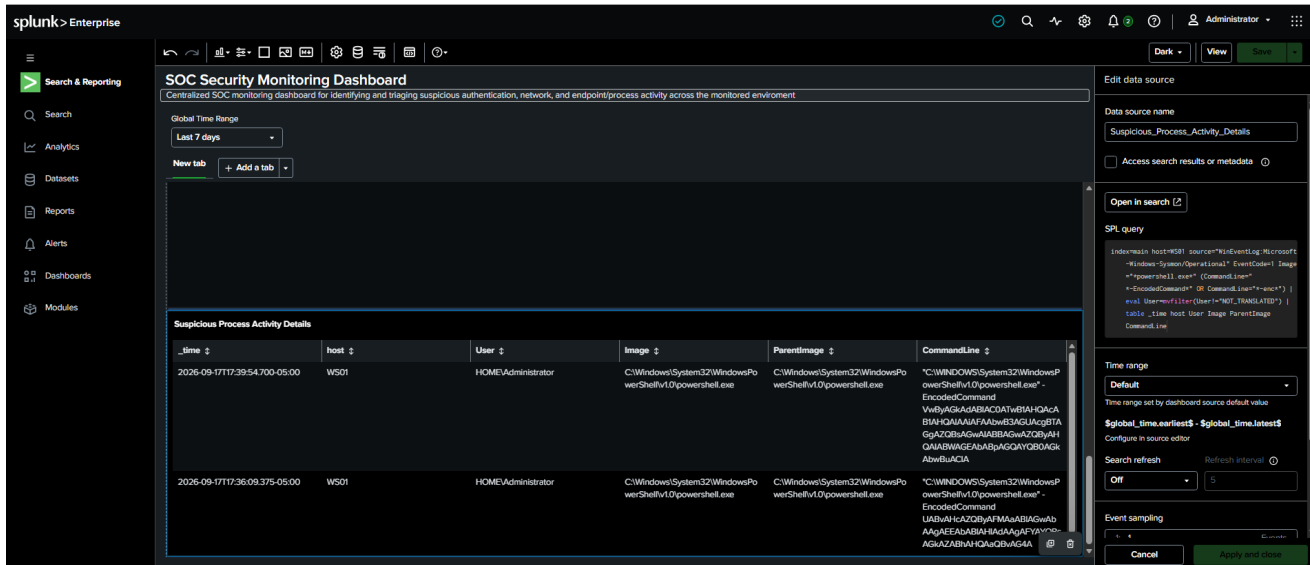


Figure 6 - Suspicious Process Activity Details showing host, user, process relationship, and encoded command-line context.

7. Dashboard Implementation & Analyst Workflow

The completed dashboard uses a global time selector and a consistent summary-to-detail layout. The top row contains the three Single Value indicators, while the corresponding full-width tables appear underneath. This design allows an analyst to identify which category contains matching activity without first reviewing large tables, then move directly to the contextual fields required for triage.

The Single Values do not represent the number of Splunk alerts that fired. They represent occurrences or grouped activity that match the criteria defined by each dashboard search within the selected time range. This distinction is important because the dashboard is a monitoring and triage interface built from telemetry and SPL, rather than a full case-management system.

8. Results

The dashboard successfully surfaced controlled activity from all three monitoring categories. The authentication search identified the intentionally generated failed-login burst from KALI01. The network reconnaissance search identified the Nmap scan from 192.168.100.30 and showed 998 unique destination ports within the qualifying two-minute window. The process search displayed two encoded PowerShell process-creation events and provided the associated host, user, Image, ParentImage, and CommandLine context.

These results demonstrated that previously developed detection concepts could be combined into a centralized monitoring view and presented in a way that supports an analyst's workflow rather than requiring each SPL search to be run and interpreted independently.

9. Skills Demonstrated & Lessons Learned

One of the biggest lessons from this project was learning which fields and results are most useful from a Tier 1 analyst's perspective. The project provided hands-on experience with Splunk Dashboard Studio, Single Value visualizations, detail tables, and the design of a monitoring interface that separates high-level indicators from investigation context.

The project also increased comfort with writing and troubleshooting SPL. Commands including stats, dc(), bin, earliest(), latest(), eval, mvfilter, where, and table were used to aggregate and present telemetry.

Troubleshooting reinforced that adding a field to a stats ... by clause changes the aggregation itself rather than simply adding another display column.

The network portion reinforced the difference between event volume and behavioral context. A large packet count does not necessarily indicate scanning, while a large number of distinct destination ports contacted in a short time window provides stronger reconnaissance context. The process portion similarly reinforced that suspicious behavior is not automatically malicious; encoded PowerShell should be investigated using process, parent-process, command-line, user, and host context before reaching a conclusion.

In a production environment, a seven-day dashboard window would not necessarily be appropriate for active monitoring. The time range and refresh interval should reflect SOC shift coverage and telemetry volume. A more mature implementation could also incorporate alert or case status so analysts can distinguish new activity from activity that has already been triaged or resolved.

10. Conclusion

This project produced a SOC monitoring dashboard in Splunk that gives a Tier 1 analyst an at-a-glance view of suspicious authentication, network, and endpoint/process activity. Three Single Value panels summarize the number of occurrences matching defined suspicious-activity criteria, while corresponding detail tables provide the contextual information needed to investigate each result.

The completed dashboard demonstrates a practical Notice -> Prioritize -> Investigate workflow and brings together authentication analysis, network reconnaissance detection, and endpoint process monitoring in a centralized interface. It also demonstrates the ability to move beyond writing individual SPL detections and consider how security telemetry should be organized and presented to support day-to-day SOC triage.